PayGuard | Production Deployment Memo

Recommendation

Proceed with a controlled, human-reviewed pilot of the Logistic Regression fraud detection prototype. Do not automatically decline or block transactions based on the model alone. The prototype is promising, but the held-out results and the gap between offline data and live banking features do not support an unsupervised production launch.

Evidence and Decision

The credit-card dataset contains 284,807 transactions, including 492 fraud cases (0.173%). Transactions were split chronologically into 70% training, 15% validation, and 15% testing. A most-frequent Dummy Classifier detected no fraud on validation (recall 0%; average precision 0.13%). Logistic Regression, with StandardScaler and balanced class weights, achieved validation average precision of 83.94%.

A 0.95 alert threshold was selected on validation data: compared with 0.50, false positives fell from 929 to 89, while recall fell from 92.86% to 87.50%. This is a starting operational choice, not a cost-optimized threshold. On the untouched test set, precision was 42.11%, recall 76.92%, F1-score 54.42%, and average precision 70.69%. The model flagged 40 of 52 fraud cases, missed 12, and incorrectly flagged 55 genuine transactions. False negatives expose PayGuard to losses; false positives create review workload and customer friction.

Proposed Production Architecture

At transaction time, PayGuard's payment service sends available transaction features to an internal scoring endpoint. The endpoint applies the saved preprocessing-and-model pipeline, returns a fraud score, and labels scores ≥0.95 for review. A rules layer routes high-risk alerts to an analyst queue; final action remains with authorized staff. Store the model version, score, decision, and review outcome in a restricted audit log. Define a safe fallback if scoring is unavailable, without treating a failed request as proof of fraud.

The dataset's anonymized V1–V28 PCA features may not exist in PayGuard's live systems, and its Time field measures elapsed time in the original dataset. Before deployment, engineering must confirm feature provenance, online availability, matching order and transformations, and no future-data leakage. Low scoring latency and consistent train/serve preprocessing are required.

Monitoring and Silent Failure

Monitor scoring latency, error and fallback rates, missing features, input distributions, alert volume, and review-queue backlog. Once delayed fraud outcomes arrive, track precision, recall, false-positive rates, missed fraud losses, and performance by relevant transaction segments. Watch for silent failures such as valid-looking but misordered features, changes in transaction patterns, or declining fraud detection despite healthy API uptime. Set investigation triggers and retraining/recalibration reviews; log model and threshold versions for auditability.

Privacy, Residency, and Retention

Keep transaction features and scoring logs within the required in-country environment. Encrypt data in transit and at rest, restrict access by role, and avoid storing full card identifiers or unnecessary personal data. Agree retention periods for raw events, model logs, and analyst decisions with legal/compliance; apply deletion and access-audit controls. No customer transaction data should be sent to an external model service without approved contracts and residency review.

Phased Rollout and Dependencies

Phase 1: Validate live feature availability, privacy controls, and historical backtesting with Risk, Engineering, and Compliance. Phase 2: Run in shadow mode, generating scores without affecting payments; compare alerts with later fraud outcomes. Phase 3: Introduce a limited analyst-reviewed pilot, measuring review capacity, losses, and customer impact. Phase 4: Expand only after agreed operational and safety targets are met; recalibrate the threshold using real costs. Dependencies include payment-platform integration, reliable fraud labels, a review team, observability, security approval, and an incident/rollback owner.

Decision Requested

Approve a shadow-mode evaluation and a small human-reviewed pilot, contingent on feature feasibility, in-country data handling, and jointly defined success and rollback criteria. Defer automated blocking until prospective evidence demonstrates an acceptable trade-off between prevented fraud and customer harm.